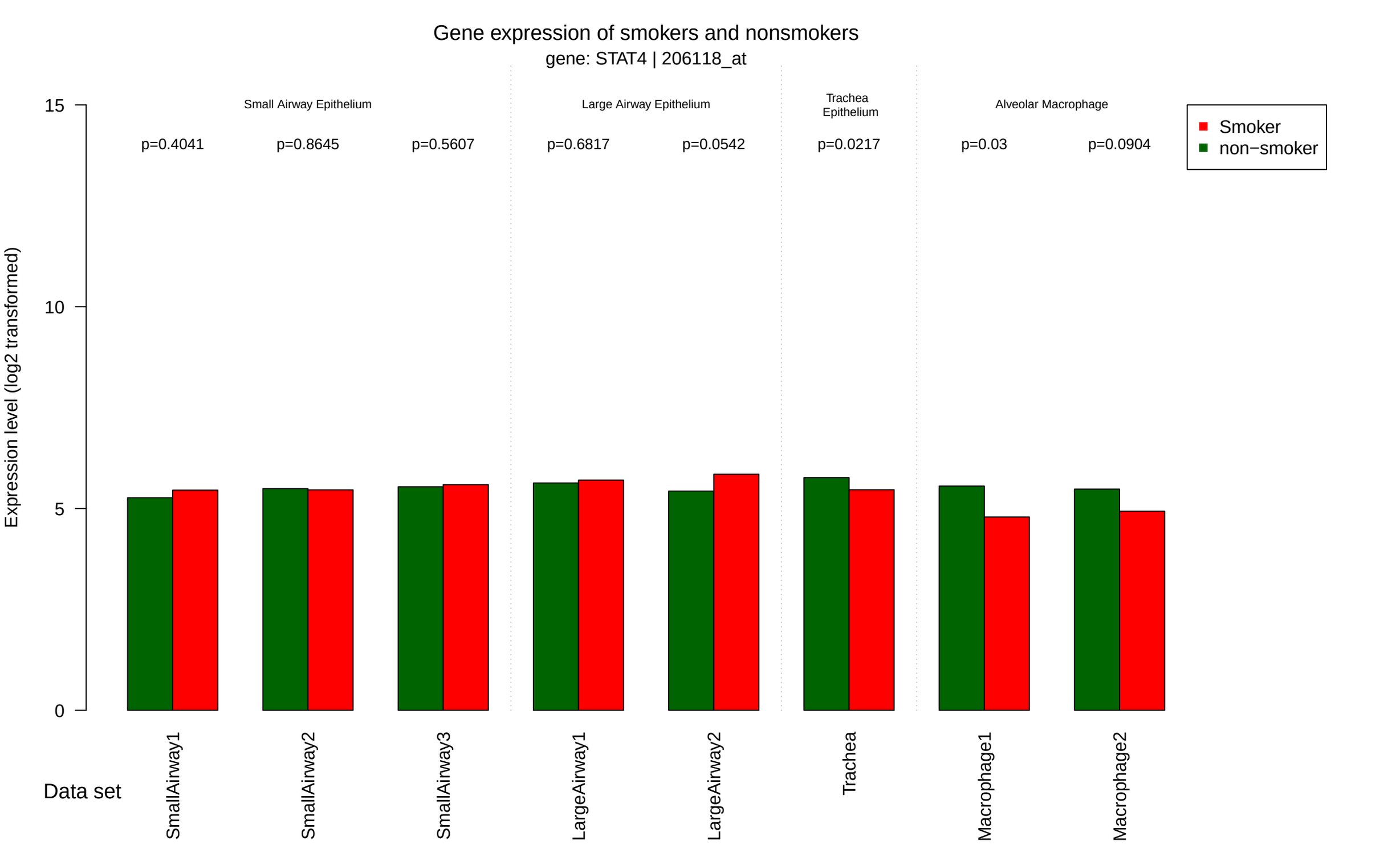
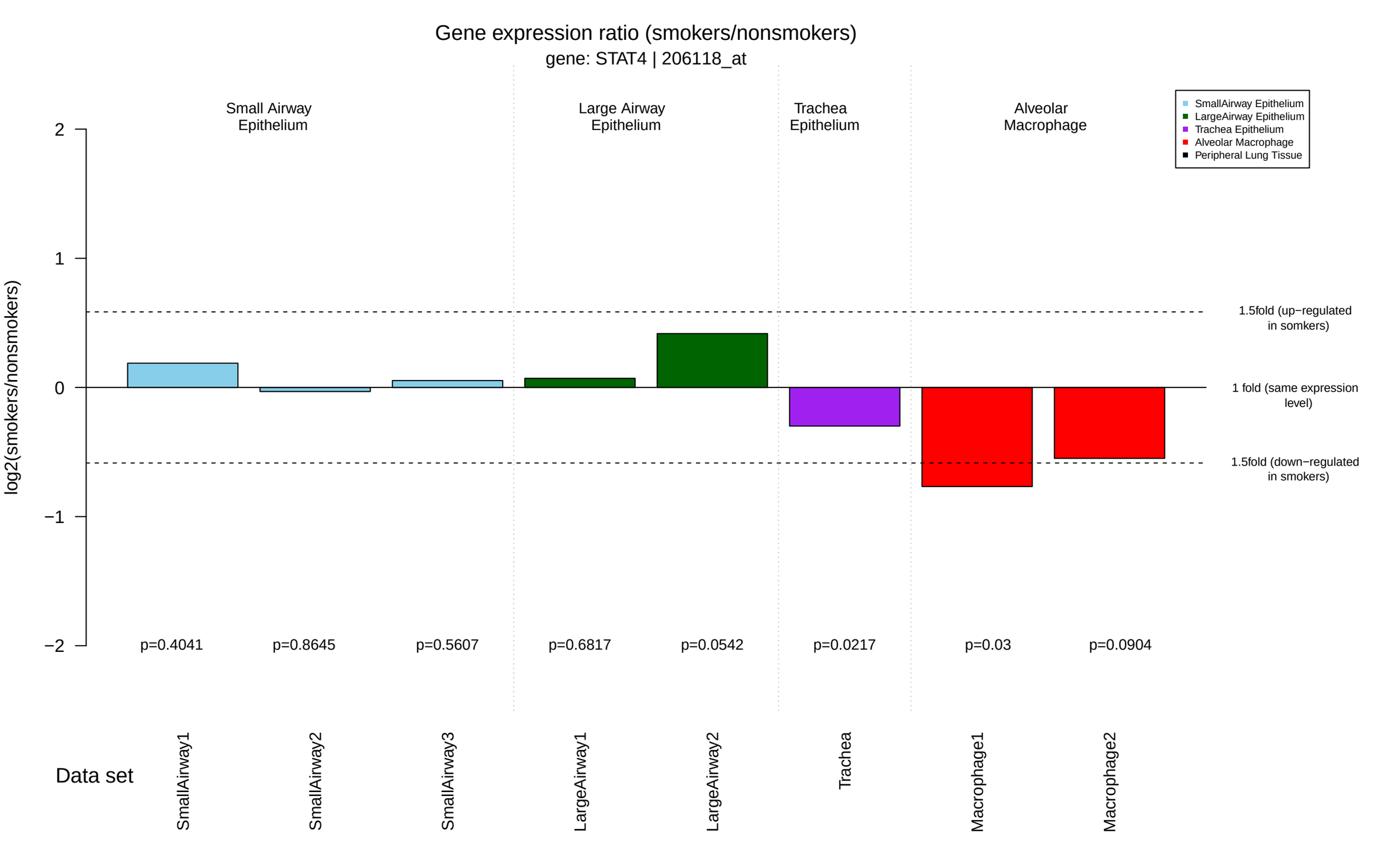
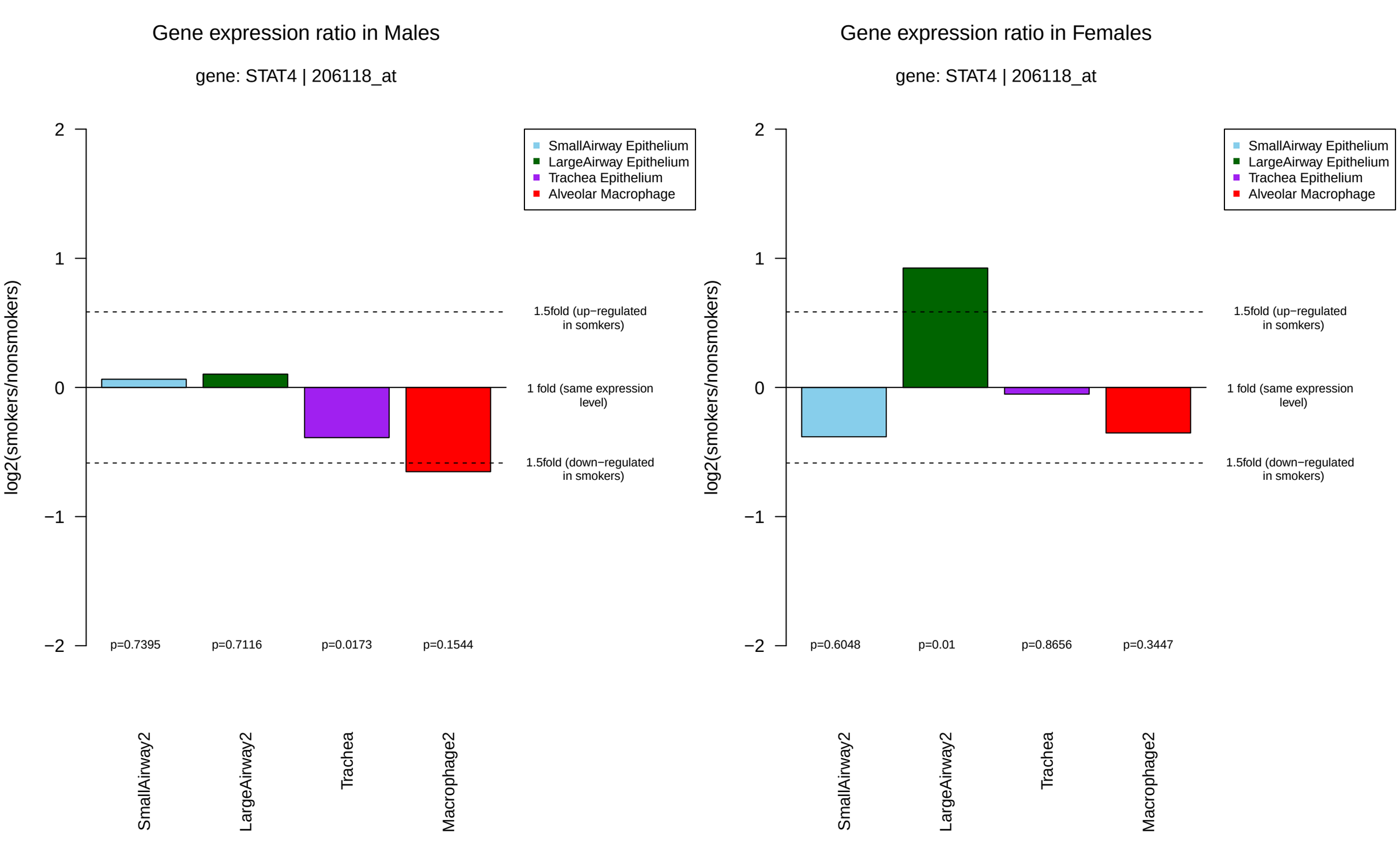
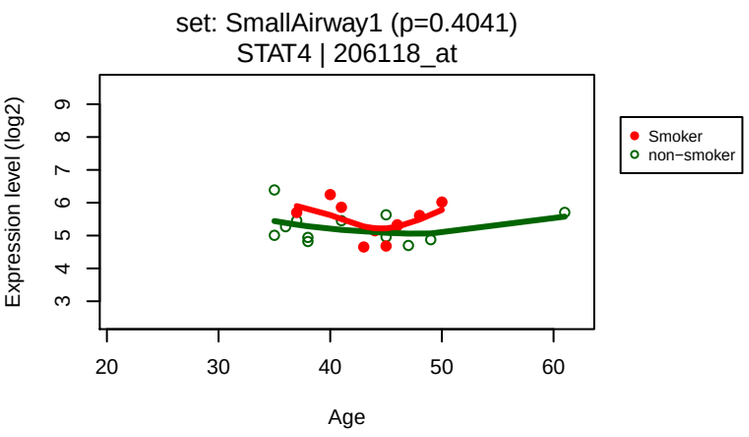
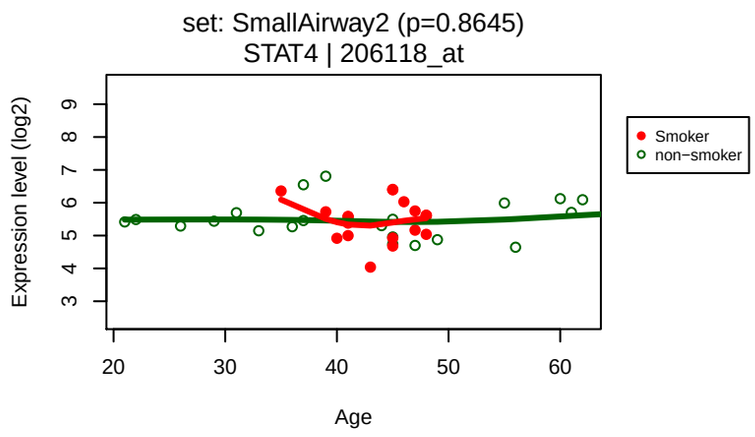
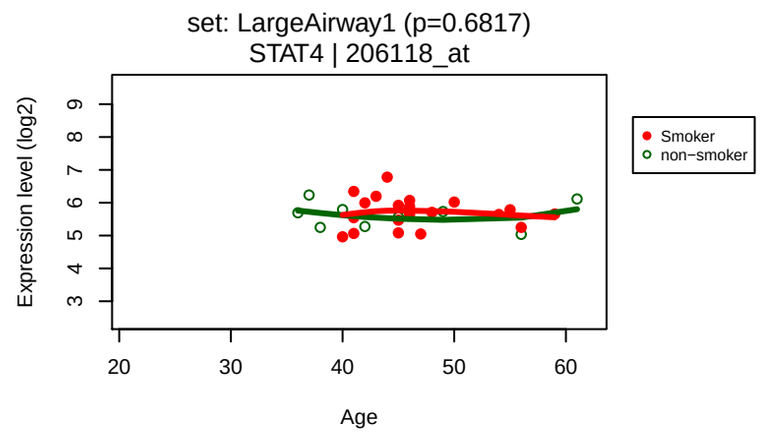

Gene expression of smokers and nonsmokers
gene: STAT4 | 206118_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4041
p=0.8645
p=0.5607
p=0.6817
p=0.0542
p=0.0217
p=0.03
p=0.0904
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

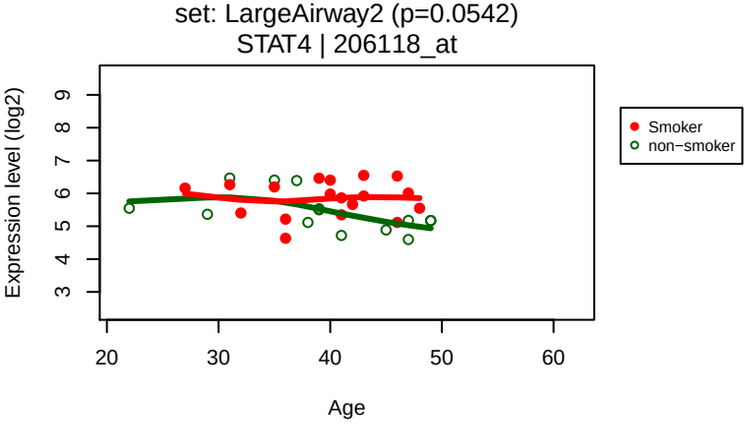

Gene expression ratio (smokers/nonsmokers)
gene: STAT4 | 206118_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4041
p=0.8645
p=0.5607
p=0.6817
p=0.0542
p=0.0217
p=0.03
p=0.0904
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

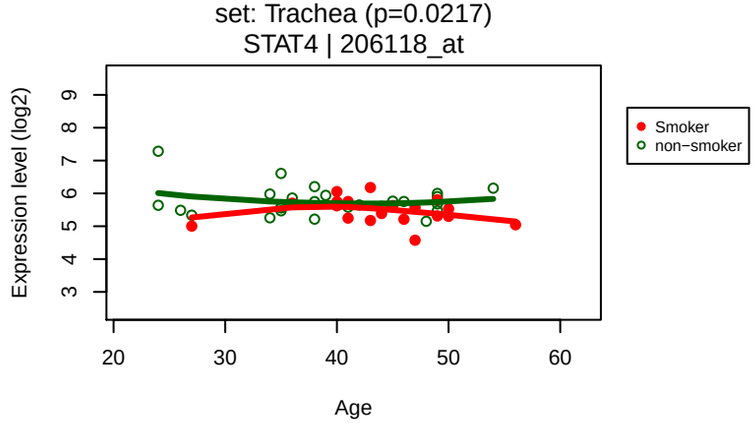

Gene expression ratio in Males
gene: STAT4 | 206118_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7395
p=0.7116
p=0.0173
p=0.1544
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT4 | 206118_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6048
p=0.01
p=0.8656
p=0.3447
SmallAirway2
LargeAirway2
Trachea
Macrophage2

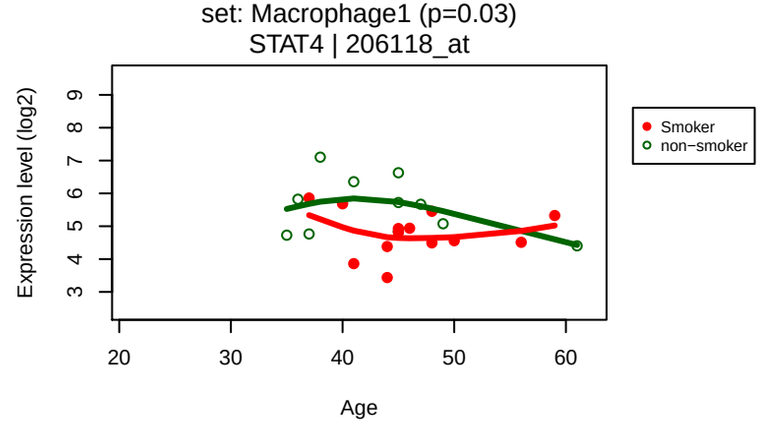

set: SmallAirway1 (p=0.4041)
STAT4 | 206118_at
Expression level (log2)
Age
Smoker
non−smoker

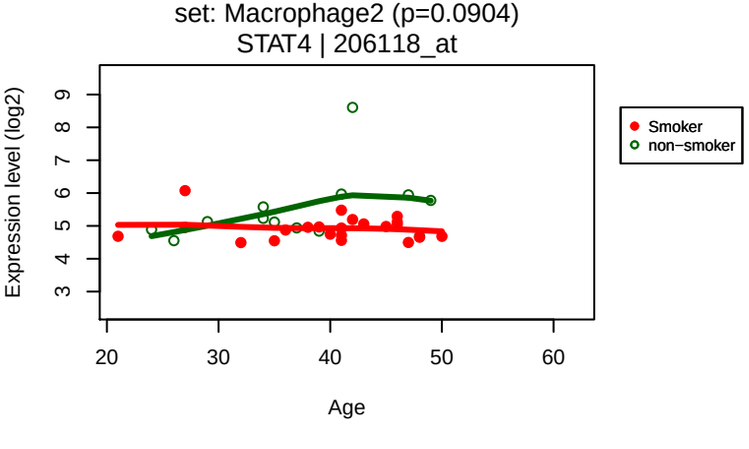

set: SmallAirway2 (p=0.8645)
STAT4 | 206118_at
Expression level (log2)
Age
Smoker
non−smoker

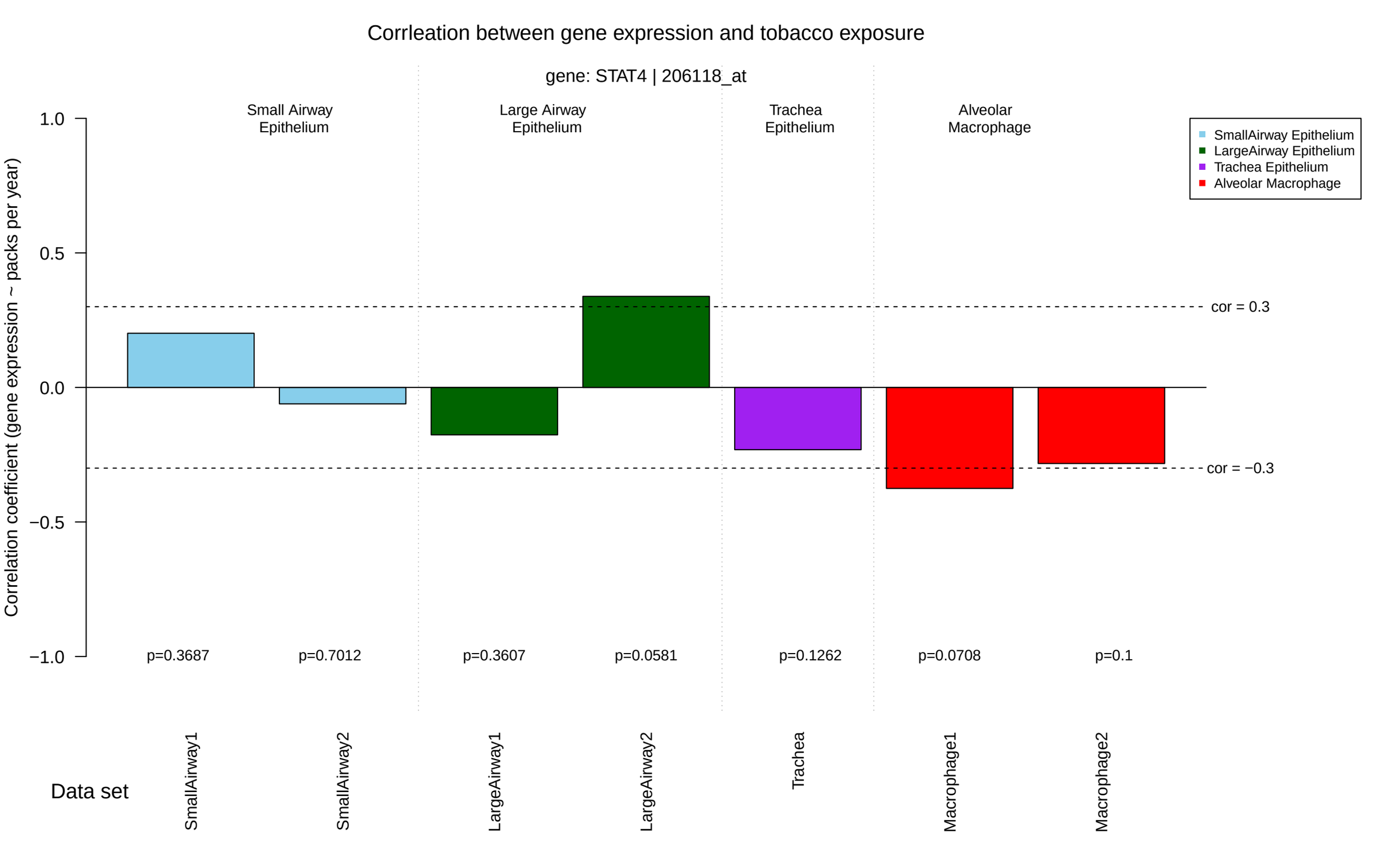

set: LargeAirway1 (p=0.6817)
STAT4 | 206118_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0542)
STAT4 | 206118_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0217)
STAT4 | 206118_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.03)
STAT4 | 206118_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0904)
STAT4 | 206118_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT4 | 206118_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3687
p=0.7012
p=0.3607
p=0.0581
p=0.1262
p=0.0708
p=0.1
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2